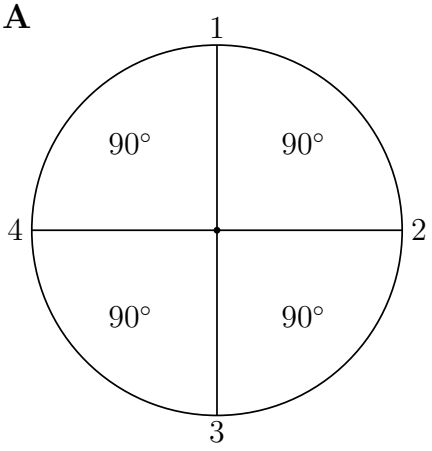
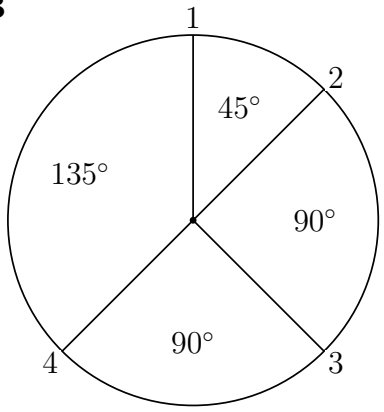
Use the same marked front for all mountain (M) and valley (V) labels. Fold every listed ray fully flat, with no extra creases or cuts. Layers may touch but may not pass through one another. A failed attempt alone does not establish impossibility.

Problem 1: Decide which patterns can close flat. Record a successful fold with M/V labels for each one you can make; leave any undecided pattern marked with a question mark.

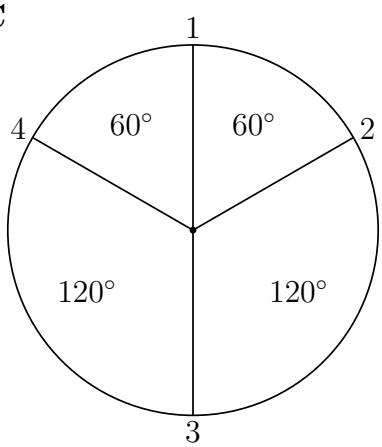
A



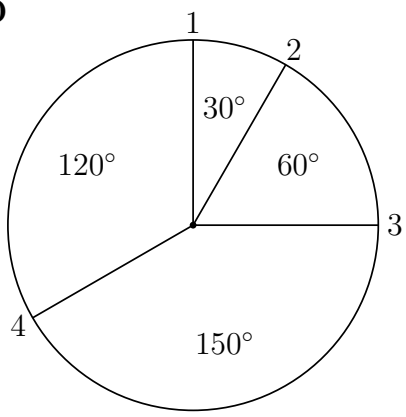
B



C

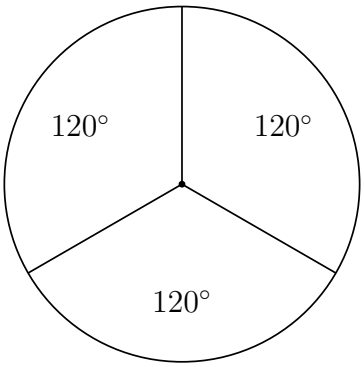


D

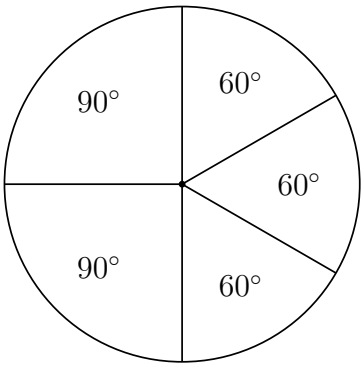


Problem 2: Can any of these patterns close flat along every ray? Explain an answer that covers all three patterns.

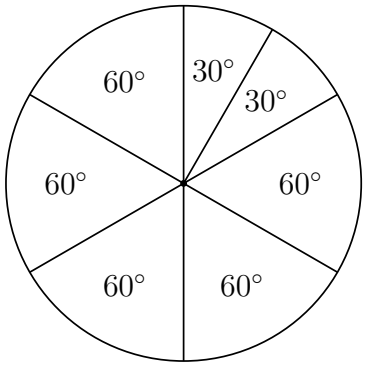
A



B

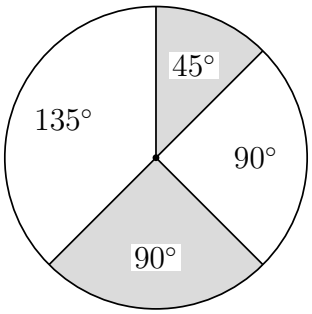


C

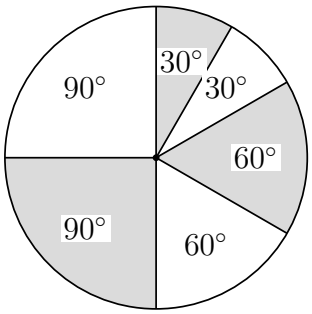


Problem 3: Explain why every other sector must total 180° in any flat-folding pattern with an even number of rays. Which patterns does your explanation rule out?

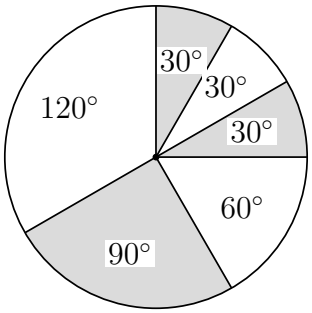
A



B



C



D

